

VARSHITHA THILAK KUMAR

Aspiring AI/ML Engineer

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EDUCATION

Amrita Vishwa Vidyapeetham – Ettimadai, Coimbatore
B-Tech – Computer Science & Engineering (AI) CGPA: 8.6 / 10

Aug 2023 – Present

Velammal Vidyalaya – Chennai Grade: 87.2 / 100
Maharishi Vidya Mandir – Chennai Grade: 93.6 / 100

July 2023

July 2021

EXPERIENCE

Wezume – AI/ML Intern
Remote, India

Jan 2026 – Present

- Built an ML culture-fit scoring model that extracts vocal pitch features from audio and ranks candidates against recruiter profiles — integrated into Wezume's live hiring platform via REST API.
- Owned full pipeline: audio feature extraction → model training → API deployment, reducing manual candidate screening effort significantly.

PROJECTS

Oil Spill Drift Forecasting | PyTorch · CNN-LSTM-Attention · SAR · Geospatial | [GitHub](#)

2026

- Designed a CNN–LSTM–Attention deep learning model to forecast oil spill trajectories from multi-modal satellite inputs (SAR imagery, wind, ocean currents, bathymetry) with physics-based advection constraints.
- Built an end-to-end train–evaluate–visualize pipeline with custom physics loss terms and CI/CD via GitHub Actions.

Battery Data Extraction – LLM + XAI Pipeline | Groq Llama-3-70B · SHAP · Scikit-learn · Streamlit | [GitHub](#)

2025

- Engineered an NLP pipeline using Llama-3-70B (Groq) + tabula-py to extract 8 battery metrics from unstructured scientific PDFs — achieving ~90–95% extraction accuracy.
- Trained a Random Forest Regressor on document features; deployed a Streamlit dashboard with SHAP-based explainability and CSV export.

Movie Inferno – AI Recommendation Platform | Next.js · Supabase · TMDB API · AI Chatbot · Vercel | [GitHub](#)

2025

- Shipped a full-stack movie recommendation app with an integrated AI chatbot, real-time Supabase sync, and production deployment on Vercel.

SKILLS

Languages: Python, C++, JavaScript, SQL, Bash
ML / DL: PyTorch, TensorFlow, Scikit-learn, Keras, OpenCV, Hugging Face Transformers
LLM / NLP: LangChain, RAG, OpenAI API, Groq (Llama-3-70B), ChromaDB, FAISS, pdfplumber
Deployment: REST APIs, Docker, Vercel, GCP, Firebase, Supabase, Streamlit, Gradio
Tools: Weights & Biases, SHAP (XAI), Git, MATLAB, Next.js, PyTorch Geometric

ACHIEVEMENTS & CERTIFICATIONS

- IEEE Conference Paper – ETIS 2024 (Trivandrum): Time-Frequency Analysis of Power Quality Disturbances, published on IEEE Xplore | Automated Battery Materials Data Extraction from Scientific Literature using LLMs and XAI (Reccap-2026 IIT Palakkad)
- Finalist – GenAI Hackathon, Value Health Org (2024) | Finalist in HackBLR (2026) | Finalist – Cubicle Code 5.0 Hackathon (2025) at Microsoft ,Bangalore
- Oracle Cloud Infrastructure 2025 Certified Foundations Associate | Multimodal RAG (GenAI Academy) | Full-Stack MERN (IITM Research Park)